

Decision on JCGS-23-236.R1

Journal of Computational and Graphical Statistics <onbehalf@manuscriptcentral.com>

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To: Emily Anna Robinson <erobin17@calpoly.edu>

 2 attachments (57 KB)

Attached standard file: * AuthorsSubmissionGuidelinesS1-for-upload-to-S1.pdf; Attached standard file: * AuthorsSupplGuidelinesS1-for-upload-to-S1.pdf;

09-Apr-2024

Dear Dr Robinson:

Thank you for submitting the manuscript "Perception and Cognitive Implications of Logarithmic Scales for Exponentially Increasing Data: Perceptual Sensitivity Tested with Statistical Lineups" for possible publication in the Journal of Computational and Graphical Statistics.

A referee, an Associate Editor (AE), and I have reviewed your paper. We agree that the paper has been improved, but some issues remain. As such, I cannot accept your paper in its current form. Nonetheless, given its potential, I would consider a revision that adequately addresses all of the reviewers' comments.

I appreciate your patience as we work to provide high quality feedback. Please don't hesitate to reach out if you have questions.

Articles published in JCGS are ordinarily accompanied by online publication of the data and computer code used to illustrate the proposed methods. The goal is to enable readers to replicate much of the analysis and simulations described in a paper and to apply proposed methods in their own data without having to write code from scratch. When it is not practical to include a whole dataset in the manuscript, the manuscript should state how the complete dataset can be obtained. Please let me know if you would not be able to publish your data for reasons of security or confidentiality. Please provide the computer code and data that you hope to publish along with your revision. I have attached the instructions for including online materials here.

Any appendices should be removed and uploaded as a separate file.

Please bear in mind that your revision is due four months from today. A revision sent later than the deadline may be considered a new submission. When you provide your revision, please include a point-by-point response to each report in a single, blinded, attachment. When uploading your response file, please choose the "Author Response to Reviewers" file designation from the drop-down menu. Please be sure to blind your response.

Please also have each author take the time to login to the system and use the Edit Account link (located in the upper right corner) to enter all of their detailed information (assuming they did not do so at some time previously).

Regarding color images: I encourage you to provide figures that can be printed in gray-scale but can be viewed electronically in color. If your paper is ultimately accepted and you wish your figures to be

printed in color, we may need to ask your assistance to offset the additional cost of printing. Details and a payment form will be provided if needed. For an estimate please contact the production editor at Rebecca.Corpier@taylorandfrancis.com.

When you are ready to submit your revision, please use the link below.

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This link will direct you to the first page of your revised manuscript.

This link will remain active until you have submitted your revised manuscript. If you begin a revision and intend to finish it at a later time, please note that your draft will appear in the "Revised Manuscripts in Draft" queue in your Author Center.

IMPORTANT: Your original files are available to you when you upload your revised manuscript. Please delete any redundant files before completing the submission.

Upon submitting your revision, you will be asked if the manuscript has been submitted to the journal previously. Click 'Yes' and then cut-and-paste the manuscript number from the email you received.

Because we are trying to facilitate timely publication of manuscripts submitted to Journal of Computational and Graphical Statistics, your revised manuscript should be uploaded by 07-Aug-2024. If it is not possible for you to submit your revision by this date, we may have to consider your paper as a new submission.

Once again, thank you for submitting your manuscript to Journal of Computational and Graphical Statistics. I look forward to receiving your revision.

Sincerely,

Faming Liang

Editor, Journal of Computational and Graphical Statistics

Reviewer(s)' Comments to Author:

Reviewer: 1

Comments to the Author

I reviewed this manuscript previously. As I said last time, I think there is some worthwhile information in this study. I was pleased that many of my and the other reviewer's comments have been incorporated. For example, moving the lineup example to the methods makes a lot more sense, and the way it is described is much clearer. However, a number of issues remain before I can recommend publication. For example,

- "In this paper, our primary focus is to evaluate the benefits and drawbacks of using log

scales, specifically delving into their impact on perceptual sensitivity towards the degree of curvature.” As I said previously, there’s nothing explicit about log scales in what the participants saw. What this does show is that people see curvature pretty well, and that for the specific functions you chose, log scales augment some curvature differences more than linear scales. In other words, this shows that you picked functions that are straighter in log space. Some text has been added to soften the conclusion, which is a step in the right direction, but my sense was that it was still framed as “are log scales better?” I suggest editing further to clarify that differences in curvature are what are being tested rather than something inherent to linear vs. log scales.

- I was pleased to see the inclusion of the Rorschach results, but would like to see them actually incorporated into the analysis. They form a great null hypothesis, so they should be used! This is particularly true given the results of the Rorschach lineups. At the bottom of page 11, the authors write that “Results from the “Rorschach” evaluations indicate null panel selections were distributed relatively evenly with multiple candidates for the most interesting panel.” In Appendix B, the authors state that [the Rorschach panel] “approach allows us to explore whether any null panels were significantly more unique, providing insights into the null plot sampling method.” The Rorschach lineup appendix shows that “the plots exhibit a relatively even distribution, offering multiple candidates for the most interesting panel. It is worth noting that in the low curvature “Rorschach” lineups simulated in data set replication one, a single panel stands out, with over 50% of participants selecting the same null panel on the linear scale and just under 50% on the log scale. Through a visual examination of “Rorschach” evaluations, we determined that our null sampling method is appropriate.” I’m not sure what to make of this. I was expecting to see a comparison against a null expectation, but instead the authors did “a visual inspection.” What does that mean? A single panel being selected ~50-60% of the time doesn’t seem like my expectation from a null model, so why does that mean the null sampling method was appropriate? In the actual test, the target panel was selected less than 50% of the time. Does that make them worse than the null? The Rorschach information could and should be used along with the results to provide more context for the results.

- I was pleased to see some results in the Participant Confidence and Accuracy appendix. However, I was hoping to see results from tests comparing the confidence levels: across accuracy (which is shown visually but not statistically) and across linear vs. log scale depictions.

- One of my earlier comments was “It would be good to see in the paper exactly how these questions were phrased and what the response options were, as these details matter for surveys.” The authors’ response was “The manuscript contains a screenshot of an example lineup shown to participants in Figure 7. We added emphasis to the left side of the screenshot which shows what participants were asked for providing their reasoning and confidence level for their selection.” Is “Which plot is the most different?” the only question asked? If so, the text needs to make that clear. If not, I suggest simply including the survey questions as supplementary material.

- On a related note, it's not sufficient to rely on a link to describe methods – they need to be in the paper.

- I asked about the “three stages of perception” indicated by the first figure. The revised manuscript includes the phrase: “Early stages of exponential growth often appear to have a small growth rate, while the middle stage seems to exhibit more quadratic growth. It is only in the later stages that the exponential growth becomes apparent.” What’s the distinction between “exhibiting quadratic growth” and “exhibiting exponential growth”? What’s the evidence for three distinct stages? I didn’t catch before the “Von Bergmann” was a reference to the comic. (Thanks to the other reviewer for catching that.) That’s not scientific evidence that there are three stages – it’s one person’s opinion crafted for humorous intent. This entire paragraph needs citations to actual peer-reviewed literature. Or, if there is no support, it should be cut.

- I still don’t understand what “determine the $y = -x$ line scaled to fit the assigned domain and range” means. The $y = -x$ line is a line; it can’t be scaled.

- Similarly, I don't understand "Map the values $x_{mid} - 0.1$ and $x_{mid} + 0.1$ to the $y = -x$ line for the two additional points."

New issues:

- "exponential model," as used in section 2.1, typically means a model without an intercept: $y = a \cdot \exp(b \cdot x)$ or something similar, without a "c" term like $a \cdot \exp(b \cdot x) + c$. The theta term in your model is the intercept, so this isn't the classic exponential model; it's a model with an exponential term.
- On a related note, the addition of the theta parameter is explained as a way to standardize the endpoints of the graph, but it also renders the main feature of log transformation (converting exponential functions to linear functions) moot.
- Make sure to cite all R packages (nlms, e.g.).
- What's the citation and/or equation for the Lack of Fit statistic?
- Fig. 5 caption: (13 should be 13
- Bottom of page 12 line 44: "Corresponding to ..."
- Page 14 line 20: Cut the comma
- Fig. 7 shows one panel (15) as grayer than the others. Presumably this was not the case when participants were choosing?
- I don't know this journal's editorial preference for including discussion in the results section. If it's acceptable, fine, but otherwise move the discussion-y parts of the results to the discussion.
- Page 22 line 45: I don't see any thumbnail figures on the right.
- I don't know this journal's preference, but I'm used to seeing a Discussion section include some references to put the results in the context of the existing literature.
- Appendix A1: Make sure to point out that this function (exponential function, which does not have an intercept) is much different than the function shown in the experiment.

Editor's Comments to Author:

Associate Editor

Comments to the Author:

(There are no comments.)

RW